ALERT SYSTEM IMPLEMENTATION

Performance Metrics Report

HealthTrack: Remote Monitoring for Home-Based Cardiac Rehabilitation

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Summary

All measurements below were taken on the development machine (Apple silicon Mac mini, 8 cores, macOS 26.5.1, Python 3.12.13, SQLite 3.53.3) by `scripts/run_reports.sh` on November 12, 2026. Requests run in-process, so network time is excluded. Sections 1 to 8 give every measurement with its sample size; the raw values are in `reports/alert_performance.json` and `reports/performance.json`.

Requirement or goal	Target	Measured	Result
NFR-07: critical alert on the dashboard after ingestion	5 s at p95	14.3 ms (alert event), 14.7 ms (notification) at p95	Met
Cost of alert evaluation on ingestion	Small next to the request	+0.36 ms per reading at the median (3.43 ms on, 3.07 ms off)	Met
NFR-08: read endpoints, with alerting active	300 ms at p95	Slowest endpoint 12.74 ms at p95 (21-day daily summary)	Met
Alert fatigue control	Far fewer notifications than threshold crossings	150 crossings produced 8 alerts and 8 notifications (95% fewer); 93 suppressed, 5 repeats folded	Met
Notification dispatch	Sent without waiting for the poll interval	6.23 ms at p95 from queued to sent	Met
Scheduled jobs	Finish well inside their 15-minute and 30-second intervals	Pattern detection 0.34 s for 105 patients; escalation check 39 ms for 108 open alerts	Met
Live stream fan-out	Many dashboards per API process	About 250,000 event deliveries per second at 10 and 50 subscribers	Measured

Alert system measurements

Run 2026-11-12T22:55:31+00:00 on macOS-26.5.1-arm64-arm-64bit (arm64, 8 cores), Python 3.12.13, sqlite 3.53.3. Dataset: 119,750 readings, 105 active patients. All requests run in-process, so network time is excluded.

1. Rule evaluation per reading

Measurement	n	p50 (ms)	p95 (ms)	max (ms)
Normal reading (no breach)	200	0.52	0.68	1.38
Breach folded into an open alert	200	1.24	1.36	1.56
Breach creating a new alert and notifications	200	1.48	1.70	3.15

2. Ingestion with alert evaluation on and off

Measurement	n	p50 (ms)	p95 (ms)	max (ms)
Single reading, rules on	200	3.43	4.53	9.15
Single reading, rules off	200	3.07	3.78	27.70
Batch of 500, rules on	10	69.19	84.66	93.95
Batch of 500, rules off	10	72.54	114.92	132.72

Alert evaluation adds 0.36 ms to a single reading at the median. Batch throughput: 7,227 readings/s with rules on, 6,893 with rules off. The batch medians differ by 3.35 ms, less than the spread within either set of 10 batches (up to 42.38 ms between median and 95th percentile), so this run cannot separate the cost of alert evaluation for a batch from run-to-run variation; the single-reading comparison, over 200 requests each, is the better estimate.

3. End to end: breaching reading posted to event on a connected dashboard

Measurement	n	p50 (ms)	p95 (ms)	max (ms)
Reading to live reading event	50	9.10	13.86	15.06
Reading to critical alert event	50	9.36	14.34	15.56
Reading to dashboard notification	50	9.61	14.66	15.98

Requirement NFR-07: critical alerts on the dashboard within 5 s (5000 ms) at the 95th percentile. Measured p95: 14.3 ms for the alert event and 14.7 ms for the notification. Met.

4. Notification dispatch delay (queued to sent, background workers)

Measurement	n	p50 (ms)	p95 (ms)	max (ms)
Dashboard	50	3.58	6.23	9.20
Email	50	3.58	6.23	9.20
SMS	50	3.58	6.23	9.20

Every channel for an alert is queued in the same transaction and sent in the same dispatch pass, so the channels share one delay. Email and SMS were written to the local outbox (no gateway configured), so these times exclude any external provider.

5. Scheduled jobs

- Pattern detection across 105 active patients: 0.34 s (3.2 ms per patient), 3 outcomes.
- Escalation check over 108 open alerts: 39 ms (108 escalated).

6. Alert fatigue: one noisy patient over 48 hours

A resting reading every 15 minutes with heart rate around the 100 bpm limit and systolic pressure around 160 mmHg; each alert is resolved 45 minutes after it opens.

Readings	Threshold crossings	Alerts created	Repeats folded	Suppressed	Notifications
192	150	8	5	93	8

Without deduplication and suppression each crossing would notify; here 150 crossings produced 8 notifications (95% fewer).

7. WebSocket fan-out (in-process broker)

Subscribers	Events	Deliveries	Publish cost (us/event)	Deliveries per second
1	1000	1,000	8.8	113,825
10	1000	10,000	39.5	253,007
50	1000	50,000	200.0	249,982

8. API endpoints with alert evaluation on ingest

Run 2026-11-12T22:55:45+00:00 on macOS-26.5.1-arm64-arm-64bit (arm64, 8 CPUs), Python 3.12.13, sqlite 3.53.3.

Dataset: 135,850 vital sign rows across 108 patients (loaded in 3.91 s). Requests run in-process through the full FastAPI stack (routing, authentication, authorization, validation, audit write), so network time is excluded.

Latency

Operation	n	p50 (ms)	p95 (ms)	p99 (ms)
POST /auth/token (bcrypt cost 12)	10	237.07	263.3	279.26
POST /patients/{id}/vitals (single reading)	200	3.85	4.7	6.03
POST /patients/{id}/vitals/batch (500 readings)	20	67.66	97.52	98.75
GET /patients (worklist page)	200	2.66	3.0	3.84
GET /patients/{id}/vitals (24 h page)	200	3.99	4.8	6.03
GET /patients/{id}/vitals (7 days, page 1 of n)	200	6.45	8.26	30.91
GET /patients/{id}/vitals/latest	200	5.53	6.37	6.66
GET /patients/{id}/vitals/daily-summary (21 days)	200	11.22	12.74	13.95
GET /patients/{id}/daily-activity/summary	200	2.65	2.93	3.37
POST /patients/search	200	2.92	3.96	5.98

Ingestion

Batch upload sustained about **6,715 readings per second**. Replaying the last batch returned 500 duplicates and stored nothing (54.43 ms).

Index effectiveness

Query: readings for one patient in a 24-hour window, newest 200. The patient tested has 10,521 readings; the window returned 33 of them.

	p50 (ms)	p95 (ms)
With `ix_vital_signs_patient_time`	0.04	0.04
Without the index	3.5	4.63

Speedup at p50: **87.5x**.

Plan with index: `SEARCH vital_signs USING INDEX ix_vital_signs_patient_time (patient_id=? AND recorded_at>? AND recorded_at<?)`

Plan without index: `SEARCH vital_signs USING INDEX sqlite_autoindex_vital_signs_1 (patient_id=?) | USE TEMP B-TREE FOR ORDER BY`